AutoTerm-SST: Zero-Setup Adaptive Terminology Memory for Streaming Speech Translation

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Abstract

We present AutoTerm-SST, an interactive system for terminology-aware streaming speech translation. Live domain talks contain rare entities and technical terms, yet requiring users to prepare a glossary before a session creates substantial friction. AutoTerm-SST removes this per-session glossary-setup step by routing among domain slices built offline: each streaming chunk injects at most 10 score-filtered glossary references into the prompt, normally drawn from a compact active slice, and a training-free router selects the active domain. The router combines generated target-translation topic evidence, speech-window domain probes, speech-centroid evidence, and retrieved-term metadata, without running a separate ASR or source-transcript classifier in the production path. The demo exposes retrieved terms, active slices, router decisions, and retrieval latency in a live evidence panel. Under benchmark-aligned mixed-domain inventories, automatic mode reaches 0.916 combined term accuracy, compared with 0.839 and 0.825 for fixed NLP and medicine inventories. Scale probes show warm MaxSim retrieval stays near 60–105 ms (p50–p95) for memories of up to one million terms, and an end-to-end routing probe tracks all three ACL/medicine transitions without a wrong switch. The released implementation supports both mock and live-GPU modes.

1 Introduction

Simultaneous speech translation (SST) targets technical talks, lectures, meetings, and live media. Prior SST work studies the quality–latency trade-off of prefix translation (Ma et al., 2019) and standardizes streaming evaluation (Ma et al., 2020). Recent omni speech-language models broaden the available modeling options for long-form speech translation (Qwen Team, 2025), but live systems still struggle with term-dense content: named methods, datasets, clinical terms, legal phrases, and financial entities often require specialized translations that are difficult to infer from local acoustic context alone.

Glossaries are a practical way to control terminology, but they are awkward for live use. A user may not know the exact topic before a talk starts, and asking every speaker or viewer to upload a domain glossary creates friction. A naive alternative retrieves from a broad open memory of hundreds of thousands of terms. AutoTerm-SST injects at most 10 score-filtered glossary references into each prompt; the systems problem is therefore to select a compact active inventory and rank useful terms within this cap.

AutoTerm-SST treats terminology control as an active-inventory and ranking problem. Automatic mode starts from a deployment-configured default slice, without asking the user to identify the domain, and a training-free router switches among domain slices when streaming evidence is strong. Broad open memory is used only as an optional low-confidence fallback pool.

The result is an interactive streaming SST workbench rather than a new translation model. The framework handles REST/WebSocket transport, session lifecycle, and agent routing; agents handle model inference, prompting, batching, and retrieval. The live agent uses Qwen3-Omni with vLLM continuous batching (Kwon et al., 2023), and the web and Electron interfaces expose retrieved terms, active glossary status, router decisions, and latency alongside translations.

Our contributions are:

- We build AutoTerm-SST, a pluggable streaming SST demo with a live evidence panel for retrieved terminology, active glossary state, router decisions, and latency.
- We introduce an adaptive terminology strategy that requires no user-provided glossary,

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routes among domain-specific slices, and sends at most 10 score-filtered candidates to the prompt.

- We evaluate terminology recall, retrieval latency and scale up to million-term memories, mixed-domain routing, and 32-session concurrent streaming.

2 Related Work

Streaming speech translation. SST systems must translate partial speech while balancing quality and latency. Prefix-to-prefix policies such as wait-k make this trade-off explicit (Ma et al., 2019), attention-based policies refine read/write decisions (Papi et al., 2023, 2024), SimulEval standardizes evaluation (Ma et al., 2020), and large-scale streaming models target direct simultaneous translation (Seamless Communication, 2023; Zhang et al., 2024). Speech language models make this setting more flexible (Qwen Team, 2025; Ouyang et al., 2025), but live deployment still requires session management, incremental transport, and latency-aware prompting. AutoTerm-SST focuses on adapting terminology during a stream; its contribution is a system demonstration rather than a new model architecture.

Terminology-aware translation. Terminology control has long been studied in machine translation, usually through user-provided dictionaries, constrained decoding, or training-time exposure to terminology constraints (Hokamp and Liu, 2017; Dinu et al., 2019); terms and named entities are also a known weakness of speech translation (Gaido et al., 2021). Retrieval-augmented generation offers another way to expose non-parametric memory at generation time (Lewis et al., 2020; Khandelwal et al., 2021). Many of these workflows assume that the domain or terminology is known before translation begins. RASST couples a speech-side MaxSim retriever with prompt-injected terminology (Luo et al., 2026); AutoTerm-SST builds on this mechanism by adapting the active terminology context from evidence observed during streaming.

LLM serving and demo systems. Modern LLM serving systems provide continuous batching and efficient generation APIs (Kwon et al., 2023; Zheng et al., 2024), but a streaming speech demo also needs microphone/file input, WebSocket transport, session cleanup, live evidence rendering, and mock-mode testing. Translation Canvas (Dandekar et al., 2024) targets offline translation analysis; AutoTerm-SST provides a live serving layer while keeping model-specific code in swappable agents.

3 System Overview

AutoTerm-SST uses a thin framework with swappable translation agents, while its main system contribution is the adaptive terminology loop inside the live agent (Figure 1). The framework serves the web UI, accepts REST control requests, streams audio over WebSockets, tracks session lifecycle, and routes each session by `agent_type`; it does not own model prompts, retrieval, batching, KV cache, or terminology resources. For chunk t, the agent retrieves from the current slice before decoding. The resulting streaming evidence may update the slice for a later chunk, no earlier than $t+1$, but never changes the prompt already used for chunk t.

Agent contract. Agents implement a small contract: `open_session`, `submit_audio`, `on_control`, `close_session`, `describe`, and `health`. The same UI and wire protocol support a deterministic mock agent for development, a legacy InfiniSST scheduler agent (Ouyang et al., 2025), or the AutoTerm-SST live agent built on RASST/Qwen3-Omni (Luo et al., 2026). The boundary is backend-agnostic: the live demo uses in-process vLLM (Kwon et al., 2023), and the same protocol can serve external SGLang-compatible backends (Zheng et al., 2024) without changing the terminology memory, router, or UI.

Streaming agent. For each audio chunk, the OmniAgent runs MaxSim terminology retrieval over the active slice and collects routing-only domain-probe evidence. Before decoding, the hybrid router combines current speech/retrieval evidence with target-text history from earlier chunks; an accepted switch affects only later retrieval. The agent filters and reranks candidates, caps prompt references at 10, and lets the PromptBuilder format the `term_map` before calling the in-process vLLM backend. After decoding, y_t is appended to the target history for a later routing tick. The agent emits partial translations and structured metadata, including retrieved terms, retrieval latency, active slices, prompt-reference counts, and router decisions.

Evidence protocol. The default WebSocket output remains plain text for backward compatibility. A JSON mode returns `{type, text, meta}`, where `meta.references` drives the live evidence

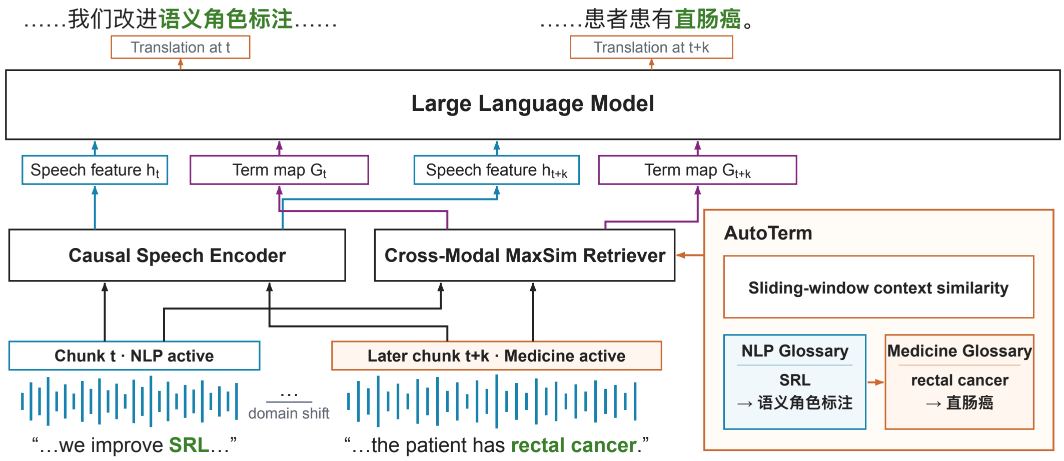

Figure 1: Architecture of AutoTerm-SST on a mixed NLP-to-medicine stream. The RASST backbone encodes each chunk and injects at most 10 bilingual term pairs from the active slice into Qwen3-Omni. Earlier, *SRL* comes from the NLP slice; after accumulated evidence confirms a domain shift, a later chunk uses *rectal cancer* from the medicine slice. The training-free router uses sliding-window context similarity to select the active glossary, and switches affect later chunks only.

panel and `meta.topic` and `meta.topic_router` describe the adaptive glossary state. This evidence protocol lets users see which terms were retrieved and whether the router stayed, switched, or deferred.

4 Adaptive Terminology Memory

AutoTerm-SST separates terminology inventory from the prompt interface. Its *terminology memory* is a collection of prebuilt inventories derived from Wikidata/Wikipedia and curated glossaries. At any time, the *active slice* is the compact domain inventory queried for each chunk; a broad *open memory* can be consulted as an optional fallback. The online system retains at most 10 score-filtered references for each prompt.

Open memory snapshots. Each snapshot maps a preset identifier to term files, metadata, and a MaxSim index. The runtime never stores large indexes in the code repository; instead, manifests point to a runtime root on the demo host. The released configuration provides NLP and medicine slices. A curated 238-term ACL glossary is used only as an evaluation oracle; the 100k open memory is an optional fallback, and the 1M memory is used for scale diagnostics.

Hybrid active glossary router. The default live router is `hybrid_window_topic`. It is not a trained topic classifier and does not run a separate ASR path or source-transcript classifier in production. Instead, it combines signals available within the streaming agent: windows of recent generated target text, routing-only speech-window domain probes, weak speech-centroid evidence, and retrieved-reference metadata:

$$\begin{aligned} score(d) =& \lambda_t \,\mathrm{topic}_d(y_{\mathrm{recent}}) \\ &+ \lambda_p \,\mathrm{probe}_d(x_{\mathrm{speech}}) \\ &+ \lambda_c \,\cos(\mathrm{EMA}(q_{\mathrm{speech}}), c_d) \\ &+ \lambda_m \,\mathrm{vote}_d(\mathrm{references}), \end{aligned} \tag{1}$$

where y_{recent} is the recent generated translation window, x_{speech} is the recent speech window used for a routing-only domain probe, q_{speech} is the pooled speech query embedding, c_d is a domain centroid built offline from a slice's MaxSim text embeddings, and reference votes come from metadata such as source preset and domain. Here, topic_d is a weighted keyword/regex score over the generated target window, while probe_d aggregates top-k MaxSim scores between the speech window and each candidate domain index. The weights and switch thresholds are hand-set: "training-free" refers only to the router and does not imply that the base model or retriever is untrained. The default weights are $\lambda_t = 0.60$, $\lambda_p = 0.25$, $\lambda_c = 0.10$, and $\lambda_m = 0.05$; they are renormalized over the signals available in a window, so routing already works before the first target text is generated. The production schedule evaluates switch decisions every 10s after a 20s warmup and caches probe scores between refreshes (Appendix A). The older embedding_refs router remains available as a compatibility policy.

Conservative switching. The router is a policy for selecting active domain slices, not an open-world topic detector. It switches only after warmup, minimum update interval, confidence threshold, top-vs-runner-up margin, repeated consistent windows, domain-probe guards, and target index readiness. A switch supported by generated target text requires multiple consecutive consistent windows; audio/probe-only switching uses stricter raw-probe floors (Appendix A). Generated target text without positive topic evidence receives no topic weight, so it cannot dilute a strong probe or turn a weak probe into a high-confidence switch. Cold index loads are non-blocking: the agent preloads a target index in the background and activates it only after the retrieval plugin reports that the index is ready.

Capped prompt candidates. Automatic mode retrieves and score-filters up to 10 candidates from the active slice. Under low-confidence fallback conditions, it may query open_wiki_100k separately, then merge, deduplicate, and rerank the two result sets. The prompt receives at most PROMPT_K $= 10$ references, with no filler terms when fewer survive. Slice size therefore trades coverage against ranking difficulty within the top-10 prompt cap. The controlled comparison uses 10k inventories as an evaluation choice, not a system limit; broader inventories remain available as fallback pools and scale diagnostics (Appendix B). The live evidence metadata records active slices, candidate-pool size, prompt-reference counts, fallback events, and retrieval-quality metrics.

5 Demo Interface

The demo is designed for three audiences: researchers inspecting terminology errors in streaming SST, lecture or conference participants who do not want to prepare glossaries, and developers testing new speech translation agents behind a stable protocol. The same interface can run in mock mode without a GPU or in live GPU mode with Qwen3-Omni and MaxSim retrieval.

Live controls. Users choose a model agent, language pair, latency multiplier, and terminology mode. In auto_working mode, the session starts from a configured default slice without requiring the user to identify the domain, and it may switch to another domain slice; every chunk uses the same top-10 prompt cap. Manual presets and ad-hoc term overrides remain available as advanced options; they affect the prompt but do not bias the router.

Evidence panel. For each partial translation, the UI renders the latest non-empty retrieved references as term-translation pairs with scores and source labels. A status line reports active memory backend, term count, and retrieval/generation latency; the adaptive row adds mode, inferred topic, confidence, active glossary, switch count, and router action. This makes terminology behavior visible rather than hidden inside the model prompt.

Packaging. The repository includes a mock-friendly launcher for UI/protocol development and a GPU launcher for the live AutoTerm-SST agent built on RASST. Mock mode exercises the same REST/WebSocket flow and JSON evidence protocol, enabling reviewers to test the demo without model weights or GPUs. The live demo additionally uses a resident vLLM engine, a dedicated retriever GPU, and an optional public tunnel.

Availability and licensing. The demo code is released under the MIT license on the GitHub repository. A source checkout is the downloadable package; in mock mode, it starts the same UI and protocol without GPUs. Live Qwen3-Omni mode requires A6000-class GPUs and the external RASST retriever checkpoint; model weights and Wikidata/Wikipedia-derived terminology resources retain their upstream licenses. Screencast and hosted-demo links are maintained on the repository demo page.[1]

6 Evaluation

Setup. We evaluate terminology memory, routing, and concurrent serving in the live AutoTerm-SST system on A6000 GPUs. Its live agent runs Qwen3-Omni with in-process vLLM tensor parallelism over two GPUs and the RASST MaxSim retriever on a third GPU. Audio is from prepared ACL 60/60 English-to-Chinese and medicine segments. Unless otherwise stated, results use the JSON WebSocket protocol, which exposes translated text, retrieved references, active glossary metadata, and per-chunk retrieval latency. A latency multiplier of 2 corresponds to 1.92s input

[1] Video and live demo: https://github.com/luojiaxuan/autoterm-sst#demo.

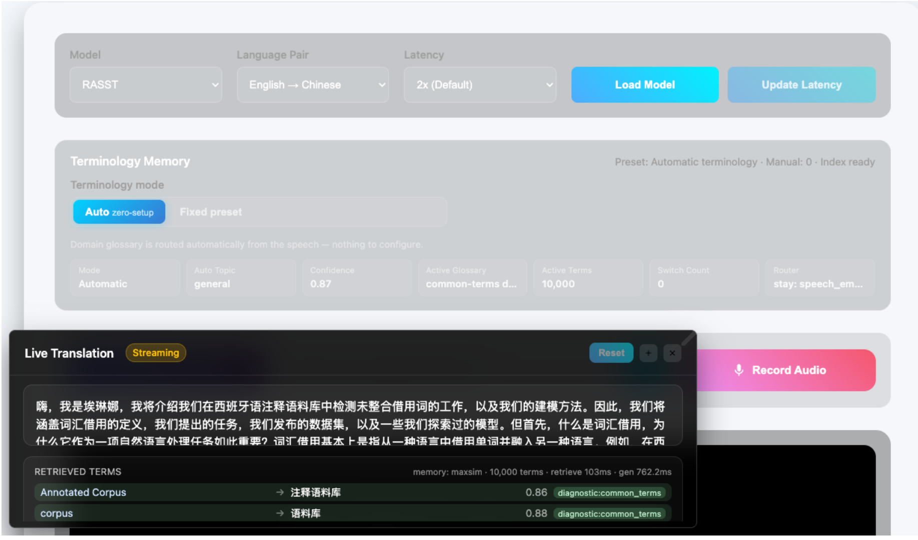

Figure 2: AutoTerm-SST web UI with live terminology evidence. The same interface controls model, latency, and terminology mode while rendering translated text, retrieved term evidence, active glossary status, router confidence, active term count, and retrieval/generation latency from the JSON WebSocket metadata. The captured session uses the diagnostic common-terms slice to illustrate the evidence schema; automatic production sessions start from domain-specific slices.

chunks. We report four complementary evaluations: a controlled mixed-domain terminology comparison, a warm-retrieval scale sweep, a mixed-domain routing probe, and a concurrent streaming stress test.

Metrics. For terminology quality, term accuracy is the fraction of annotated target-term occurrences found in the output. The controlled comparison contains 143 technical occurrences: 98 from ACL and 45 medicine occurrences spanning 23 medicine term types. Masked BLEU deletes annotated target-term strings from both hypothesis and reference before scoring, and is used only as a non-terminology sanity check. For runtime, we report prompt references per partial chunk and retrieval-latency percentiles. For routing, we report switch latency, wrong switches, and steady-state active-domain accuracy after transition grace windows. A separate 32-session run measures serving behavior under concurrent load. These metrics do not replace token-delay measures such as StreamLAAL (Papi et al., 2024).

Controlled terminology comparison. Table 1 compares equal-sized domain inventories that fully cover the gold terms in their matching domain. It therefore tests whether routing can compose domain-specific resources across a stream, rather than evaluating open-world terminology mining. Automatic mode reaches 0.916 combined term accuracy, exceeding both fixed single-domain inventories, and also has the highest BLEU in this run. Appendix C reports raw-gold, per-domain, type-level, and masked-BLEU results. Separately, a five-talk fixed-inventory sweep shows term accuracy stable at 0.965–0.979 while retrieval precision falls from 0.48 (curated) to 0.14 (100k) as distractors are added (Appendix B).

Setting	Session setup	Term acc.	ACL	Med.	BLEU
No memory	None	0.769	0.806	0.689	23.14
Fixed NLP	Select NLP	0.839	0.888	0.733	23.37
Fixed medicine	Select med.	0.825	0.776	0.933	22.92
Automatic	None	**0.916**	**0.898**	**0.956**	**24.07**

Table 1: Controlled ACL → medicine → ACL comparison under equal-sized benchmark-aligned union inventories. Each fixed domain inventory fully covers the gold terms for its matching domain; Automatic switches between the two inventories without session-specific glossary input or a domain choice. Term accuracy scores 143 occurrences (98 ACL and 45 medicine).

Memory	Terms	p50/p95 ms	Refs/chunk
Curated	238	61.3 / 105.3	2.33
Domain	10k	78.5 / 82.1	1.80
Open	100k	64.1 / 82.6	4.89
Stress	1M	78 / 95	9.62

Table 2: Warm steady-state MaxSim retrieval latency across terminology memory sizes. Cold loading grows with index size (about 0.5s for 238 terms, 6.8s for 100k, and over 30s for 1M), so AutoTerm-SST preloads candidate working indexes outside the streaming batch path.

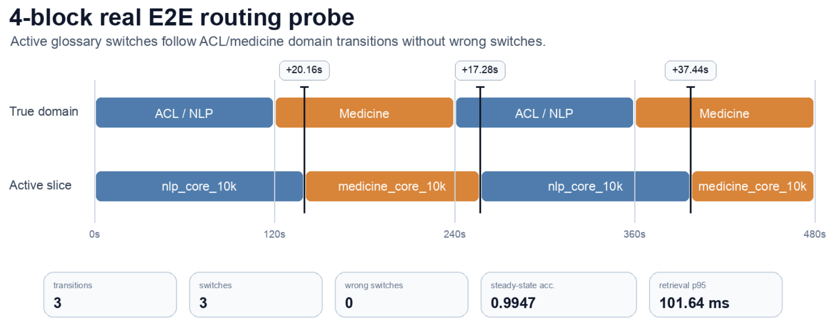

Figure 3: Mixed-domain routing probe on a 480s end-to-end streaming playlist with ACL, medicine, ACL, and medicine blocks. The automatic active glossary switches after each true domain boundary with no wrong switches; the final switch is delayed because the medicine segment begins with speaker and session framing before explicit oncology terminology appears.

Scale and prompt noise. Across Table 2, p50 ranges from 61.3 to 78.5 ms and p95 from 82.1 to 105.3 ms. Latency does not increase monotonically with inventory size in these single runs; the clearer scaling cost is cold index loading. The 100k and 1M inventories also yield more surviving prompt references than the 10k domain slice. This supports keeping the active slice compact, using the 100k open memory only as an on-demand fallback, and reserving the 1M index for scale diagnostics.

Mixed-domain routing. Figure 3 reports the four-block end-to-end routing probe for `auto_working`. The run observes three domain transitions, makes three active-slice switches, and records zero wrong switches. The first two switches occur 20.16s and 17.28s after the domain boundary; the final switch occurs 37.44s after the boundary because the opening of that medicine block contains general session framing before clear oncology terms. The run keeps the top-10 prompt invariant with 99.5% active-domain accuracy after excluding a 30s grace window following each boundary. The last switch misses the 30s target by 7.44s but is not a wrong switch. A medicine-only session started from the default NLP slice recovers by switching to the medicine slice 59.5s into the session with zero wrong switches, so the initial slice is a warm start rather than a hidden oracle. The speech probe alone is noisy (0.52 top-1 accuracy), so generated-target evidence leads routing and the probe acts as a guard. Two 640-window state-machine proxy tests, using reference-text windows and controlled probe evidence, cover 16 alternating/random transitions with zero wrong switches. Inverted- and contested-probe controls expose the expected wrong-switch and delayed-switch failure modes. These proxies are separate from the preceding end-to-end audio run. Under benchmark-aligned union inventories, automatic routing also outperforms *both* single-domain fixed inventories on a mixed playlist (0.916 vs. 0.839/0.825 combined term accuracy; Appendix C).

Concurrent streaming. In a 32-session stress test, prerecorded ACL audio was streamed to every session at its original pace (`auto_working`, 300s each). All 32 sessions completed with zero failures or disconnects; median first-output latency increased from 0.35s for one session to 0.54s for 32, and every session sustained the real-time input rate through vLLM continuous batching. This demonstrates transport and serving path under concurrent use, while the other evaluations isolate terminology-memory behavior.

7 Limitations and Conclusion

AutoTerm-SST is a live terminology-aware SST system, not a new foundation model or topic classifier. The live demo serves En→Zh/Ja/De; routing depends partly on small, hand-written target-language keyword lists, so unsupported languages or domains may switch more slowly (Appendix D). Large open memories can expand coverage but reduce retrieval precision under the 10-reference cap, so the 100k memory is queried only through a guarded fallback. The hybrid router is intentionally conservative and may retain the initial slice when evidence is ambiguous. Our evaluations are single-run system probes, and the matched-union comparison uses full-coverage benchmark inventories; neither establishes open-world or corpus-level significance. The retrieval-latency sweep excludes the additional routing-only domain probes, whose cost is recorded separately by the runtime. Finally, the real backend requires high-end NVIDIA GPUs and external model/retriever artifacts; mock mode supports UI and protocol testing without them.

We presented AutoTerm-SST, a terminology-memory workbench that requires no per-session glossary upload or domain choice. It keeps broad resources off the default retrieval path, selects active slices from streaming evidence, injects at most 10 score-filtered references per prompt, and exposes terminology, router decisions, and latency in real time.

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A Router Configuration and Guards

Table 3 lists the default production configuration of the hybrid active-glossary router as released in `configs/autoterm_slices.yaml`; deployment-specific configurations can override individual values explicitly. Guards are evaluated in order: warmup, update interval, target-equals-active, index readiness, probe floors, confidence, top-vs-runner-up margin, target-vs-current margin, post-switch cooldown, and consecutive-window consistency. Uncertain windows keep the current slice. The released configuration uses two consistent generated-target windows and a 0.30 target-vs-current margin; the two 640-window proxy tests in Section 6 used three generated-target windows.

Parameter	Default
Signal weights (renormalized per window)	
generated-target topic λ_t	0.60
speech-window domain probe λ_p	0.25
speech centroid λ_c	0.10
reference metadata λ_m	0.05
Switch thresholds	
activation confidence	0.60
top-vs-runner-up margin	0.15
target-vs-current margin	0.30
consistent windows (text / gen. target / audio)	2 / 2 / 3
Audio-probe floors / schedule / retrieval	
probe top score / margin / positive domains	0.50 / 0.08 / 2
warmup / update / cooldown / staleness	20s / 10s / 30s / 120s
cap `PROMPT_K` / score threshold	10 / 0.78

Table 3: Default hybrid router configuration. Active inventory slices are typed: two configurable domain slices (NLP and medicine), a 100k open-memory fallback pool, a curated 238-term oracle (evaluation only), and a diagnostic common-terms slice.

B Glossary-Scale Streaming Sweep

Table 4 reports the five-talk streaming sweep referenced in Section 6: five concatenated ACL 60-60 talks (3,442s of audio) streamed through the live JSON WebSocket at latency multiplier 2, scored against a curated 142-term technical gold set. Each inventory always contains the full 238-entry curated ACL glossary; larger rows add Wikidata-mined AI/CS candidates as distractors. These are manual-preset runs isolating inventory scale (not `auto_working` runs). Term accuracy, BLEU, and masked BLEU stay in a narrow band up to 100k terms while retrieval precision collapses, motivating compact routed active slices.

Inventory	Terms	Term acc.	BLEU	M-BLEU	Gold@10	Prec@10	Refs
Curated ACL	238	0.972	58.27	53.10	0.972	0.480	1.53
+AI transl. 10k	10,238	0.965	58.72	53.35	0.972	0.234	3.09
+AI broad 10k	10,238	0.972	58.38	52.75	0.979	0.367	2.00
+AI broad 50k	50,238	0.979	58.65	53.24	0.979	0.196	3.64
+AI broad 100k	100,238	0.965	58.83	53.32	0.979	0.137	5.03

Table 4: Five-talk streaming sweep (latency multiplier 2) against the 142-term technical gold set. Gold@10 is the fraction of gold terms surfaced within the top-10 retrieval cap; Prec@10 is retrieval precision; Refs is surviving prompt references per chunk. At latency multiplier 1 the same pattern holds (term accuracy 0.944–0.958; precision 0.479 → 0.133).

C Mixed-Domain Terminology under Union Inventories

A glossary-channel comparison is only meaningful when the gold terms are inside the retrieval inventory: the broad wiki slices cover 29/142 of the ACL technical gold and 1/54 of the `medicine_606` gold, so their term accuracy mostly measures base-model retention. We therefore evaluate with *benchmark-aligned union inventories* at a fixed 10k inventory size: the byte-identical gold glossary (238 ACL / 212 hard-medicine terms) plus seeded wiki fillers, mirroring the recipe of Appendix B (gold coverage 142/142 and 54/54).

Table 5 is a controlled full-coverage comparison on a mixed ACL → medicine → ACL playlist (600s per block, live streaming at latency multiplier 2). Automatic mode requires no session-time glossary input or domain choice but relies on inventories built offline. It exceeds *both* single-domain fixed inventories on the combined gold, switching twice with zero wrong switches (37.4s / 25.0s after the boundaries; steady-state active-domain accuracy 1.0; retrieval p95 101.6 ms).

D Multilingual Terminology (En → Ja/De)

The terminology resources provide three target-language translations: the gold glossaries carry zh/ja/de values, a single MaxSim index serves every target language (translations are resolved from the term list at prompt time), and the language pair selects the per-language demo model. Table 6 repeats the mixed full-length ACL → medicine → ACL comparison for En→Ja and En→De. Automatic routing, with no session-time glossary input, has the best combined term

Setting	Acc_{tech}	Acc_{raw}	ACL	Med.	Med. types	BLEU	M-BLEU
None	0.769	0.779	0.806	0.689	9/23	23.14	20.96
Fixed NLP	0.839	0.867	0.888	0.733	11/23	23.37	20.32
Fixed medicine	0.825	0.819	0.776	0.933	20/23	22.92	20.29
Automatic	**0.916**	**0.912**	**0.898**	**0.956**	**21/23**	**24.07**	20.49
Gold-only medicine	0.811	0.801	0.776	0.889	18/23	22.94	20.59

Table 5: Mixed-domain streaming comparison with union inventories. Acc_{tech} scores 143 occurrences (98 ACL + 45 medicine), and Acc_{raw} scores 226 (181 ACL + the same 45 medicine); the latter 45 occurrences span 23 medicine types. Per-domain columns and M-BLEU use the technical gold. "Gold-only medicine" retrieves from the 212-term medicine resource without seeded fillers. Automatic routing outperforms both fixed single-domain inventories on combined term accuracy in this controlled run.

accuracy in every language under both gold denominators, with zero wrong switches and steady-state active-domain accuracy 1.0. With the released ja/de topic keywords, switches land within 29–52s (ja 28.9/33.5s; de 52.0/33.5s; zh 37.4/25.0s). An ablation without language-tuned keywords degrades first-switch latency to 115.3s (ja) and 875.6s (de), when routing leans on the slower probe channel. These delays coincide with lower medicine coverage and BLEU while switch directions stay correct; the keyword lists are small, high-precision regex sets per domain. German term matching is case-folded with light stem tolerance (exact matching undercounts under inflection); short variants keep strict word boundaries.

	Setting	Acc_{tech}	Acc_{raw}	ACL	Med.	Med. types	BLEU	M-BLEU
Ja	None	0.703	0.720	0.730	0.687	25/53	43.55	41.60
	Fixed NLP	0.718	0.760	0.900	0.608	21/53	44.19	42.21
	Fixed medicine	0.861	0.837	0.720	**0.946**	**48/53**	**44.90**	**42.83**
	Automatic	**0.910**	**0.900**	**0.910**	0.910	45/53	40.10	38.01
De	None	0.745	0.740	0.740	0.747	24/49	37.52	37.47
	Fixed NLP	0.762	0.768	0.840	0.720	23/49	37.21	36.91
	Fixed medicine	0.840	0.812	0.670	**0.934**	**43/49**	**37.63**	**37.18**
	Automatic	**0.911**	**0.880**	**0.890**	0.923	41/49	36.97	36.17

Table 6: En→Ja/De mixed-domain comparison (full-length playlist; ja: 266/350 technical/raw gold occurrences, 53 medicine types; de: 282/366, 49 types). Automatic rows use the released configuration with ja/de topic keywords. Automatic routing has the highest combined term accuracy in both languages, while its BLEU remains below the fixed-medicine setting.